Lecture 10 Homework

Programming

1. Write and test code for the Naor-Reingold Pseudorandom Number Generator.
2. Write and test code for the Blum-Blum-Shub Pseudorandom Number Generator.

Exercises

1. Consider the Generalized Linear Congruent Generator (LCG) $L(x_n) = x_{n+1} = 7x_n + 2 \bmod 13$, where the seed is $x_0 = 6$. What is this LCG's periodicity? Does the periodicity depend on the seed? Find the value of the of $x_n \ni L(x_n) = x_n$.
2. Consider the Generalized Linear Congruent Generator (LCG) $L(x_n) = x_{n+1} = 6x_n + 9 \bmod 11$, where the seed is $x_0 = 4$. What is this LCG's periodicity? Does the periodicity depend on the seed? Find the value of the of $x_n \ni L(x_n) = x_n$.
3. Consider the Generalized Linear Congruent Generator (LCG) $L(x_n) = x_{n+1} = 7x_n + 2 \bmod 11$, where the seed is $x_0 = 4$. What is this LCG's periodicity? Does the periodicity depend on the seed? Find the value of the of $x_n \ni L(x_n) = x_n$.
4. Consider the Linear Feed Feedback Shift Register (LFSR) defined by recursion $s_{n+1} = s_n - s_{n-1}$ where s_i is a Real number. Show that for any $\text{seed} = (s_0, s_1)$ the periodicity is 6.
5. What is the period of Feed Feedback Shift Register (LFSR) defined by recursion $x_{n+1} = x_n + x_{n-1} + x_{n-2} + x_{n-3}$ where $\text{seed} = (x_0, x_1, x_2, x_3) = (0,1,0,0)$?

Suggested Reading

- Garrett – Chapter 21
- Smart – Section 5.6
- Forouzan – Appendix K